Frequency-Severity GLM Pricing Model for Motor Third Party Liability Insurance

1. Purpose of the Project

This project was undertaken to deepen my understanding of General Insurance pricing techniques and enhance my practical experience in actuarial modelling within the motor insurance market.

The project covers a simple end-to-end actuarial pricing model for a Motor Third Party Liability (MTPL) insurance portfolio using Generalised Linear Models (GLMs). While this project follows a standard frequency-severity modelling framework commonly used within general insurance pricing, claim frequency and claim severity are modelled separately before being combined to estimate policy-level pure premiums and derive rating relativities.

2. Data

2.1 Data Source

The analysis utilised the French Motor Third Party Liability (MTPL) dataset, sourced from the online resource platform Kaggle (MTPL Dataset).

The dataset consists of two files;

File	Granularity	Key Fields
freMTPL2freq.csv	One row per policy	IDpol, ClaimNb, Exposure, VehPower, VehAge, DrivAge, BonusMalus, VehBrand, VehGas, Area, Density, Region
freMTPL2sev.csv	One row per claim	IDpol, ClaimAmount

The two files are linked through IDpol which is renamed to policy_id during data preparation and validation stage.

2.2 Data preparation

Prior modelling each dataset was validated and errored records were removed. The datasets were validated under the following criteria;

2.2.1 Frequency dataset

- Duplicate policy records: the additional record under the same policy_id were removed
- Negative claim counts were removed
- Exposure values outside the range (0,2] were removed
- Driver ages outside the range [18,100] were removed
- Vehicle ages less than zero were removed and capped at 20 years
- Bonus-Malus score outside the range [50,350] were removed.
- Vehicle power was winsorized to range [4,15]
- Population density was log transformed as the values panned across larger magnitudes
- Case for vehicle fuel type was standardised
- New ‘has_claim’ column created as an indicator variable to identify policies with at least one claim

Frequency dataset set originally had 678,013 rows and 2 rows with errors were found and removed.

2.2.2 Severity dataset

- Claim amounts negative or zero were remove

- Extreme claim amounts were capped at the 99th percentile which is €16,794

Severity dataset consist of total 26,639 rows and no errors were found.

The two datasets were joined through the use of policy_id and average severity for each policy were calculated (total claims / claim count) and included in the final joined dataset named 'modelling_data'.

2.3 Other data adjustments

Bands were created for continuous variables as follows;

- drv_age_band - Driver age, banded into 8 groups (18–22 through 71+)
- veh_age_band - Vehicle age, banded into 5 groups (0–1 through 11+)
- veh_power_band - Vehicle power banded into 5 groups
- bm_band - Bonus-Malus score, banded into 8 groups (50–54 through 151+)
- density_band - Log population density, cut into 5 equal-exposure quantiles

Reference levels were set for each variable, the most populated band/ category was selected as the reference level.

The full modelling data set was split to train data and test data at 70:30 proportion. The split was stratified on has claim indicator.

2.4 Exploratory data analysis output

All the exploratory data analysis were performed on the training data to ensure no test data was exposed to the model.

2.4.1 Portfolio summary

Following is a summary of the training dataset.

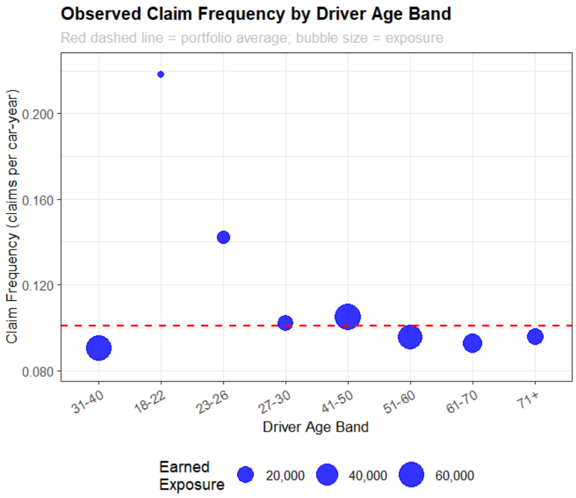
- Total earned car-years : 250,946
- Total claim count : 25,296
- Observed claim frequency : 0.1008
- Mean claim severity : €2,360
- Total incurred claims : €28,733,125

2.4.2 Observed claim frequency by rating factor

For each variable (rating factor) the claims per car year was plotted against band/category. Bubble plots were used to show the earn exposure simultaneously.

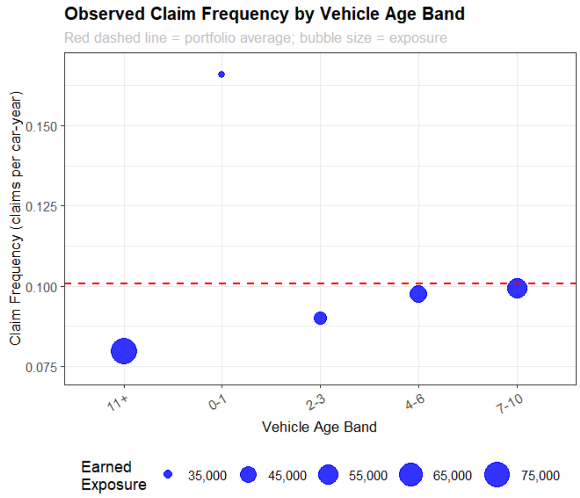
Following results were obtained;

Driver Age



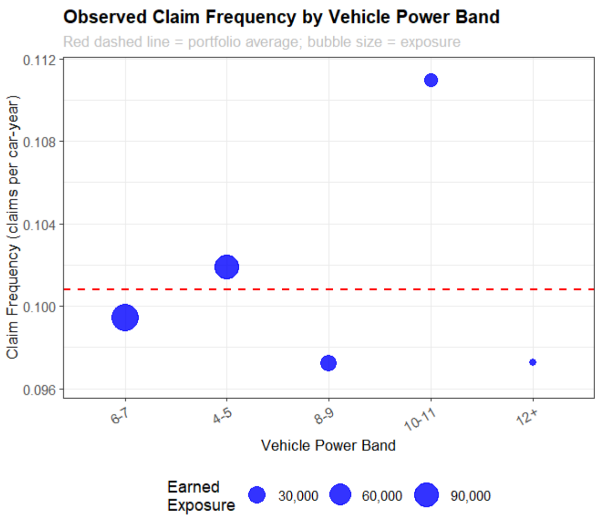
Young drivers (18–22) show claim frequency nearly double the portfolio average, which sits around 0.10. Frequency drops sharply through the 20s and 30s, reaching its lowest point in the 41–50 and 51–60 bands, before rising again for older drivers (71+)

Vehicle age



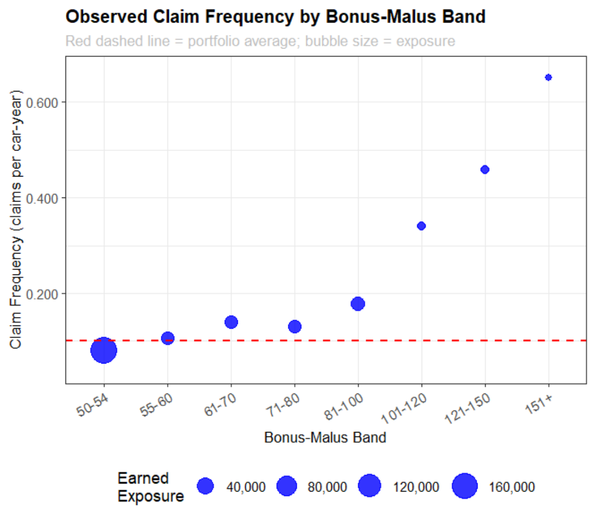
Newer vehicles (0–1 years) have noticeably higher claim frequency than older ones, with frequency declining steadily as vehicle age increases.

Vehicle power



Frequency difference among the bands is relatively weak and highest frequency observed in the band 10-11.

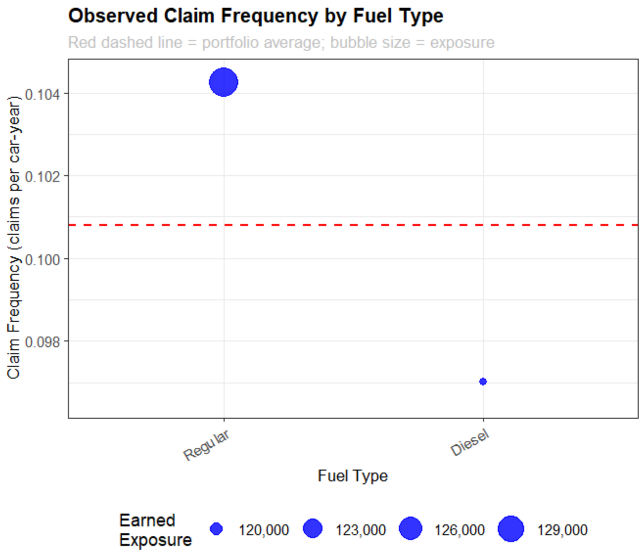
Bonus-Malus Score



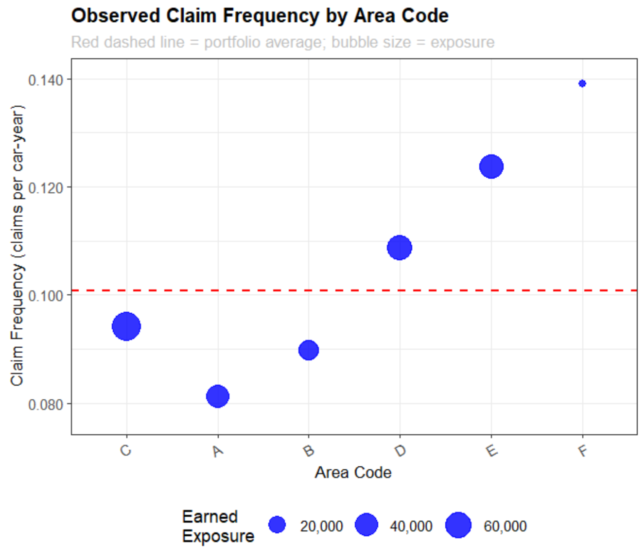
Frequency rises exponentially with the bm score which is expected as it reflects prior claims experience.

Vehicle Fuel type

Area Code

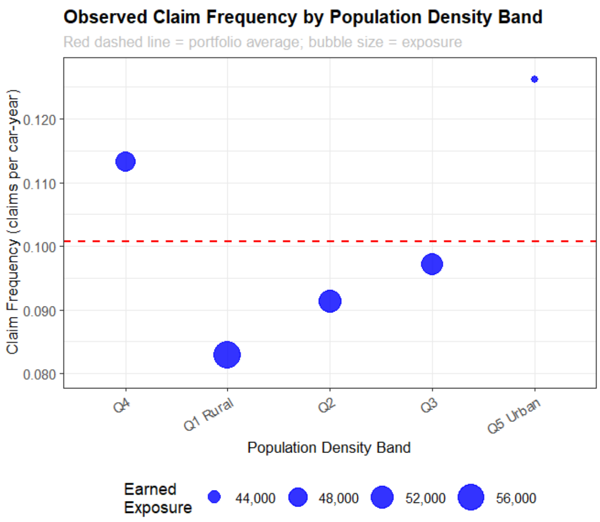


No significant difference between the frequencies however exposure for diesel vehicles is significantly low.



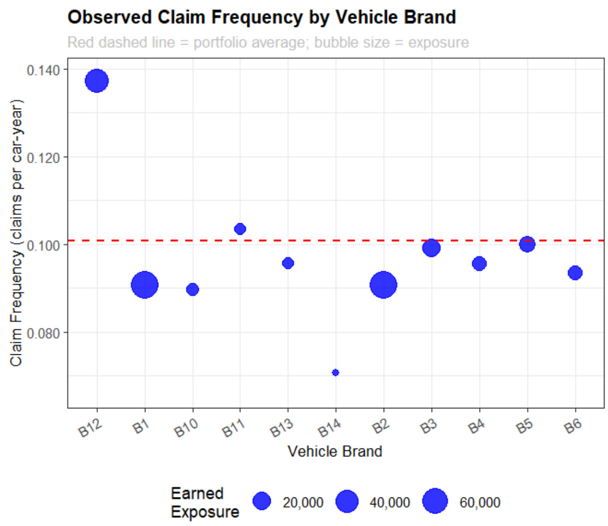
A clear geographic gradient. Area A (rural) to Area F (urban and highly populated) steady increase in the claim frequency as expected.

Population Density



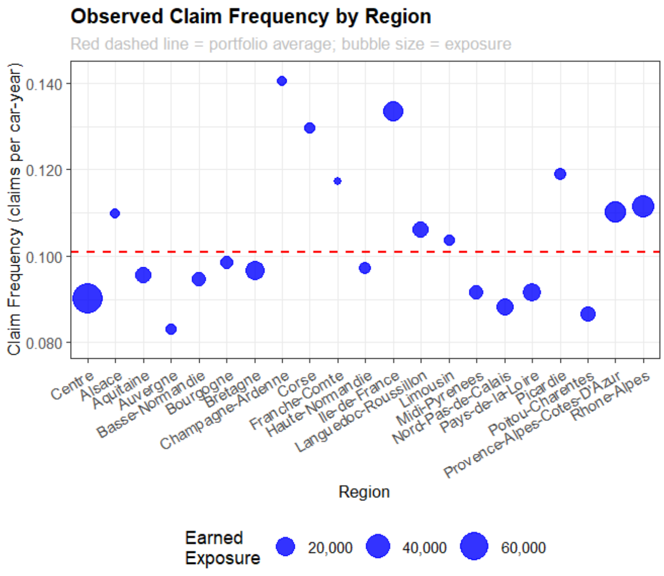
A clear positive relationship, denser areas (Q5 Urban) have higher claim frequency than rural areas (Q1) as expected.

Vehicle Brand



Some variation across brands, with B12 notably above the portfolio average and B14 notable below it.

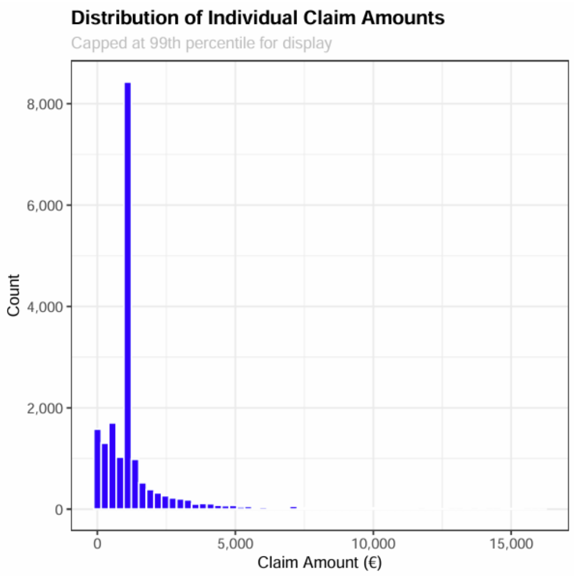
Region



Visible variations across different regions

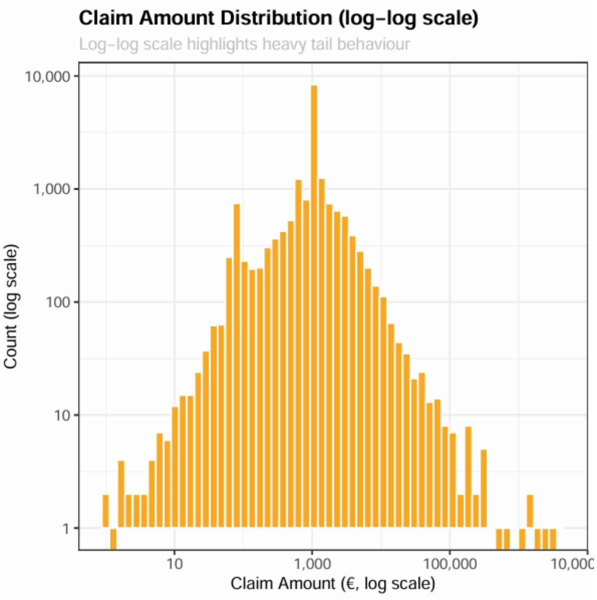
2.4.3 Severity distribution

Linear Histogram



This **Linear histogram** shows a heavily right skewed with majority claims falling below €5,000 and longtail extending past €15,000. The distribution is strictly positive and highly right-skewed, characteristics commonly associated with claim severity data and consistent with the use of a Gamma GLM.

Log-log histogram



This **Log-log histogram** distribution appears broadly unimodal with no obvious clustering, suggesting a single underlying claims process. This pattern is typical of motor insurance severity data and supports the use of a Gamma GLM for modelling claim severity.

The exposure profile created during the run confirms that the 31–40 band has the largest volume of earned car-years, lending credibility to the reference level selection, while the 18–22 and 71+ bands are thinner and their relativities should be interpreted with more caution. The exposure profiles are saved in the ‘03_EDA_Exposure_Profiles.pdf’

3. Modelling assumptions & Limitations

3.1. Assumptions

The following assumptions were made throughout the modelling process.

- Historical claims experience is representative of future experience.
- Policy exposure accurately reflects time at risk.
- Claim counts follow a Poisson distribution.
- Claim severities follow a Gamma distribution.
- Rating factors have multiplicative effects on risk.
- The selected variables capture the primary drivers of claim frequency and severity.

3.2. Limitations

- The frequency and severity models are built separately and then multiplied to get the pure premium. This assumes that how often a policyholder claims and how much they claim are unrelated which might not always be true
- The dispersion checks in section 6 shows an over dispersion which suggests that a Negative binomial model maybe more suitable than the Poisson model.
- The severity is modelled as an average cost per policy rather than at individual claim level. Even though this keeps things simple some details about how claims are distributed within a policy is lost.
- Variable selection for GLMs are done using backward selection at this initial stage however it does not guarantee the best possible set of variables.
- No interactions between variables are being tested at this stage. In motor insurance some combinations can be more material for an example a young driver with high power car can be riskier than the two factors alone
- The banding boundaries were chosen using judgement and a basic credibility check.
- The large loss cap at the 99th percentile is based mainly on the data. In practice reinsurance structure will be considered.

4. Model Approach

Frequency and severity were modelled separately, and a frequency-severity framework was adopted to obtain expected claim cost

$$Expected\ Claims\ Cost\ =\ Claim\ Frequency\ \times\ Claim\ Severity$$

4.1. Claim frequency

Claim frequency was modelled using a Poisson GLM with log link function where the target variable was Claim count per policy. Log(exposure) was included allowing the model to estimate claim frequency rather than raw claim count.

$$log(Expected\ claims)\ =\ log(Exposure)\ +\ X\beta$$

Candidate rating factors included all available variable expect for region. Stepwise selection using AIC was used to identify and remove any variables that do not improve the model fit. Final frequency GLM was selected to be;

$$claim\_count \sim drv\_age\_band\ +\ veh\_age\_band\ +\ veh\_power\_band\ +$$
$$bm\_band\ +\ veh\_fuel\ +\ density\_band\ +\ region\ +\ veh\_brand\ +$$
$$offset(log(exposure))$$

The frequency model coefficients, including standard error, P value, z value, upper and lower confident intervals for each coefficient is printed to the 'freq_model_coefficient.csv' output file.

Predicted frequency for each policy is calculated and included in the final version of the modelling data table in R under 'pred_freq' column.

4.2. Claim Severity

As previously mentioned, the claim severity was modelled using the Gamma GLM with log link and was weighted by claim count for credibility. The target variable for this model is the average severity.

Candidate rating factors included all available variables. Stepwise selection using AIC was used to identify and remove any variables that do not improve the model fit. Final severity GLM was determined to be;

$$avg\_severity \sim drv\_age\_band\ +\ veh\_age\_band\ +\ veh\_power\_band\ +\ region +$$
$$area$$

The severity model coefficients, including standard error, P value, z value, upper and lower confident intervals for each coefficient is printed to the 'sev_model_coefficient.csv' output file.

Predicted severity for each policy is calculated and included in the final version of the modelling data table in R under 'pred_sev' column.

4.3. Expected Claim Cost (Pure Premium)

Predicted annual frequency (pred_freq) and severity (pred_sev) values were multiplied to calculate policy level pure premium (pred_pp), which was stored in the modelling data table within R.

$$Pure\ Premium\ =\ Expected\ Frequency\ \times\ Expected\ Severity$$

The resulting pure premium estimates provide a risk-based measure of expected claim cost and form the basis for premium differentiation across the portfolio.

5. Model diagnostics & Validations

5.1. Frequency model

5.1.1. Residual analysis

The following diagnostics were obtained on the Frequency fitted model;

- Null deviance : 157450.2 on 474606 df
- Residual deviance : 151500.8 on 474569 df
- Deviance dispersion : 0.319
- Pearson dispersion : 2.613
- Anova summary :

```
Model: poisson, link: log

Response: claim_count

Terms added sequentially (first to last)


                Df Deviance Resid. Df Resid. Dev  Pr(>Chi)    
NULL                           474606     157450              
drv_age_band     7    734.9    474599     156715 < 2.2e-16 ***
veh_age_band     4   1705.8    474595     155009 < 2.2e-16 ***
veh_power_band   4     25.0    474591     154985 5.122e-05 ***
bm_band          7   3227.2    474584     151757 < 2.2e-16 ***
veh_fuel         1     60.0    474583     151697 9.257e-15 ***
density_band     4    132.1    474579     151565 < 2.2e-16 ***
veh_brand       10     64.4    474569     151501 5.240e-10 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

According to the above-mentioned statistics, the deviance dispersion shows under-dispersion which indicates residual deviance is much smaller than expected however the Pearson dispersion is a much higher value compared to the ideal 1, indicating over dispersion. The two conflicting results may have caused due to the large number of zero claim counts, model misspecification etc.

Residual vs fitted value plot (figure 1) shows most observations concentrate below zero with a clear triangular shape develops as the fitted frequencies increases. Hence the model has no severe non-linearity. The red smoother line is not entirely flat but remains close to zero. However, the widening spreads more than expected reflecting similar concerns suggested by the Pearson dispersion statistic.

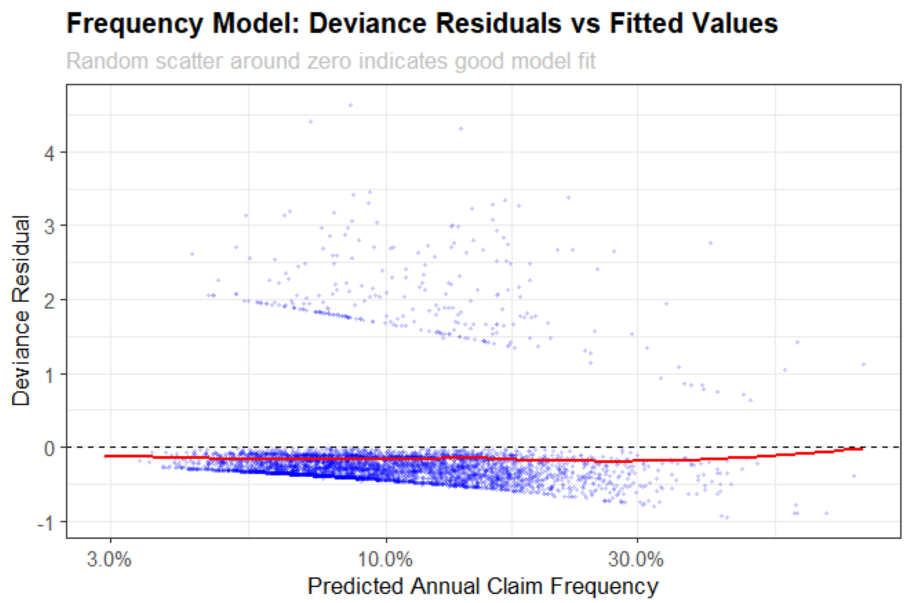


Figure 1

The **Q-Q Plot** (Figure 2) follows the straight line in the mid-section but the upper tail bends dramatically upward and the lower section bends downwards. This shows that the residuals are much heavy tailed than expected suggesting the model underestimates occurrence of unusually high claim counts. Again, pointing towards over-dispersion.

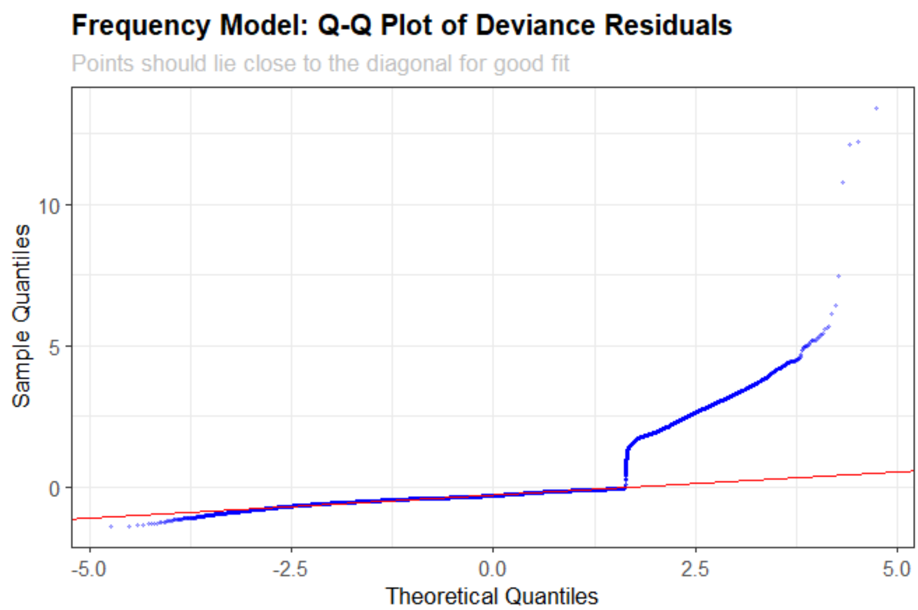


Figure 2

5.1.2. Decile Lift chart

The **Decile Lift chart** for the frequency model (Figure 3) shows policies in the test set ranked into ten groups by predicted frequency, with decile 1 being the lowest predicted risk and decile 10 the highest. The actual and the predicted lines track each other closely across all deciles which is positive.

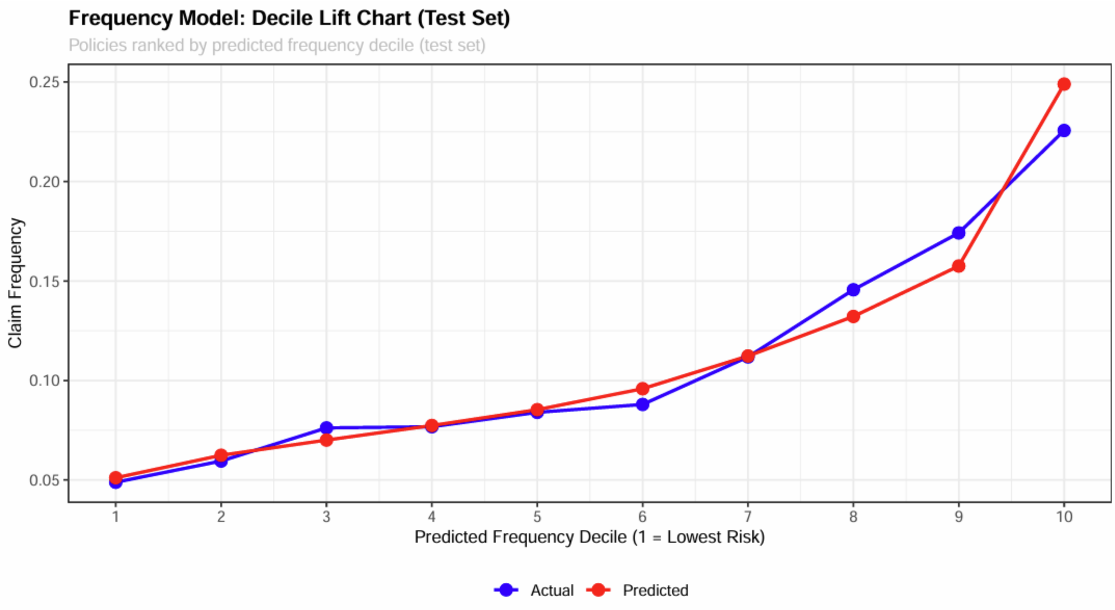


Figure 3

5.1.3. Conclusion

The frequency model provides a reasonable fit to the data however evidence of over-dispersion suggests that the model require further investigation, especially on the use of Negative Binomial distribution for the GLM.

5.2. Severity model

5.2.1. Residual analysis

The severity model itself has considerably less rating factor compared to the frequency model.

The following diagnostics were obtained on the Severity fitted model;

- Null deviance : 18207.8 on 17421 df
- Residual deviance : 17940.7 on 17381 df
- Gamma dispersion (phi) : 1.9283
- ANOVA :

```
Analysis of Deviance Table

Model: Gamma, link: log

Response: avg_severity

Terms added sequentially (first to last)


                Df Deviance Resid. Df Resid. Dev  Pr(>Chi)
NULL                            17421      18208
drv_age_band     7   53.059     17414      18155 0.0002689 ***
veh_age_band     4   59.485     17410      18095 3.286e-06 ***
veh_power_band   4   18.178     17406      18077 0.0512644 .
area             5   33.235     17401      18044 0.0040737 **
region          20  103.125     17381      17941 6.903e-05 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

The Gamma dispersion is much higher than the ideal 1 suggesting that the substantial variability in claim size is beyond what the available rating factor could explain. Same conclusion derived from the residual and null deviances.

The **Residual vs Fitted Values** (figure 4) show a substantial scatter with a negative trending red smooth line. This indicates system bias, indicating the predictions are being shrunk towards the mean.

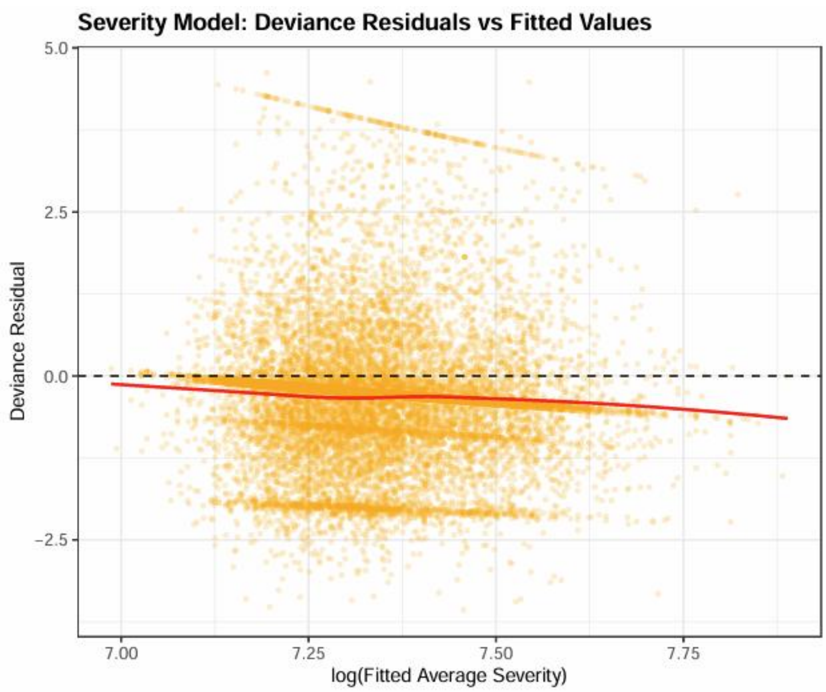


Figure 4

Q-Q Plot (figure 5) indicates an S-shape which suggests that the residual distribution is far from what Gamma GLM assumes. The model is failing to reproduce very small and very large claims.

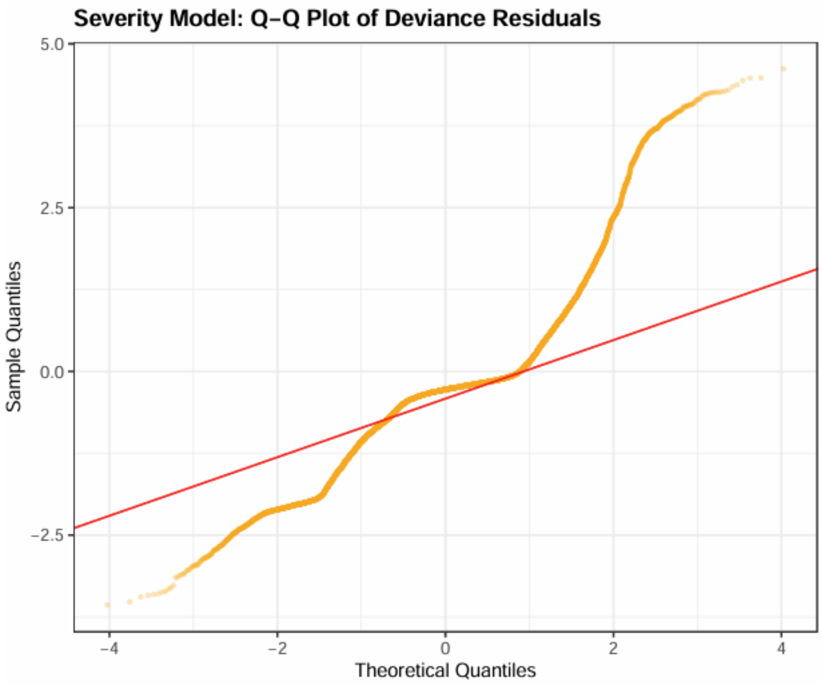


Figure 5

5.2.2. Decile Lift chart

The **Decile Lift chart** (figure 6) for the severity model shows policies in the test set ranked into ten groups by predicted claim amount, with decile 1 being the lowest predicted risk and decile 10 the highest. The actual and the predicted lines track each other more closely at the later section while at the lower deciles the predicted claim amount seems to be much lower than the actual. However, the model seems to have ranked the risks from low severity to high severity with general upward trend.

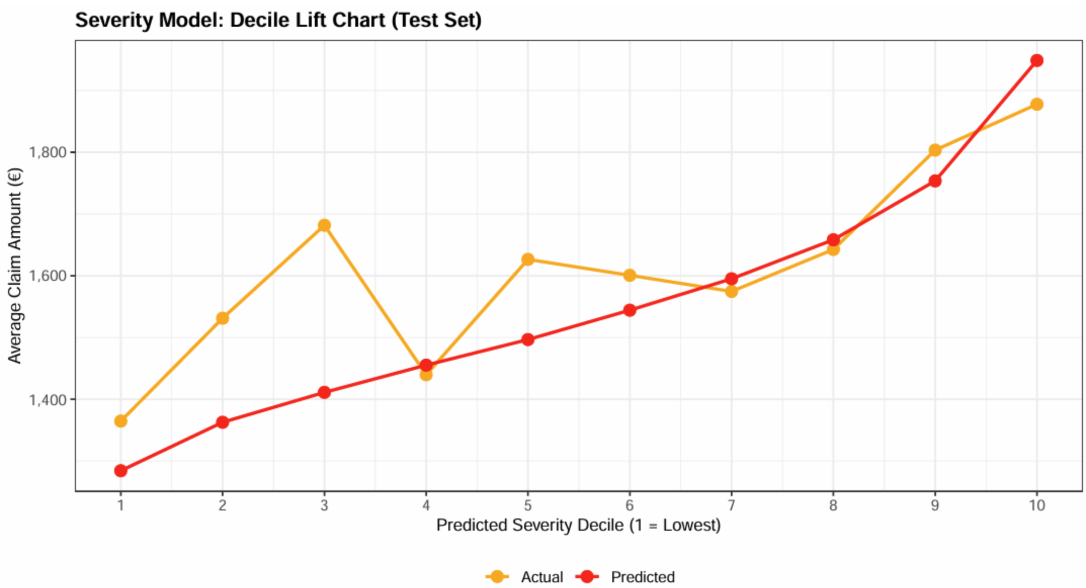


Figure 6

5.2.3. *Conclusion*

The severity model demonstrates ability to provide risk differentiation across policy holders but fails in predicting individual claim amounts accurately. Further investigations might be necessary on additional claim-level variable and alternate severity modelling approaches to improve the model fit.

5.3. Pure Premium

5.3.1. *Diagnostic summary*

The Pure Premium summary for this model is as follows;

- Actual frequency : 0.1007042
- Predicted Frequency : 0.100773
- Actual loss cost : 115.9997
- Predicted loss cost : 157.4545
- Loss cost ratio : 1.35737

The predicted claim frequency closely matched the observed frequency, indicating excellent calibration of the frequency component. However, the predicted loss cost exceeded the observed loss cost, producing a predicted-to-actual ratio of **1.357**. This suggests the primary source of error likely arising from the severity model rather than the frequency model.

5.3.2. *Rating Relativity Assessment*

The pure premium model indicates that Bonus-Malus, driver age and vehicle age are the primary rating factors driving expected claim costs. Bonus-Malus is the dominant predictor through its strong impact on claim frequency, while vehicle age and driver age influence both claim frequency and severity. Geographic variables primarily affect claim severity, whereas density predominantly affects frequency. Vehicle power, fuel type and vehicle brand provide only limited additional risk differentiation.

Below is a summary of key rating factors.

Rating Factor	Lowest Risk	Highest Risk	Pure Premium Range	Key Observation
Bonus-Malus	BM 50–54 (1.00)	BM 151+ (8.79)	8.8×	Strongest predictor by far

Vehicle Age	11+ years (1.00)	0–1 years (2.36)	2.4×	New vehicles have higher frequency and severity
Driver Age	27–30 (0.86)	18–22 (1.76)	2.0×	Young drivers remain highest-risk segment
Region	Auvergne (0.65)	Franche-Comté (1.32)	2.0×	Regional severity differences evident
Density	Rural (0.83)	Urban (1.02)	1.2×	Urbanisation primarily affects frequency
Vehicle Brand	B14 (0.70)	B12 (1.00)	1.4×	Modest effect
Fuel Type	Diesel (0.93)	Regular (1.00)	1.1×	Small effect
Vehicle Power	4–5 (0.90)	10–11 (1.06)	1.2×	Limited predictive power

Overall, the rating structure seems to be consistent with established motor insurance pricing theory. Additional charts are created for each variable and saved in '10_Rating_Relativities.pdf' file.

5.3.3. Distribution of Pure Premium

The **Pure Premium distribution chart** (figure 7) is positively skewed with a large concentration of policies assigned relatively low expected claims and a smaller number of policies assigned for higher pure premiums. No obvious discontinuities or extreme clustering are seen in the distribution suggesting that the model produces a smooth and reasonable spread of risk-based premiums across the portfolio.

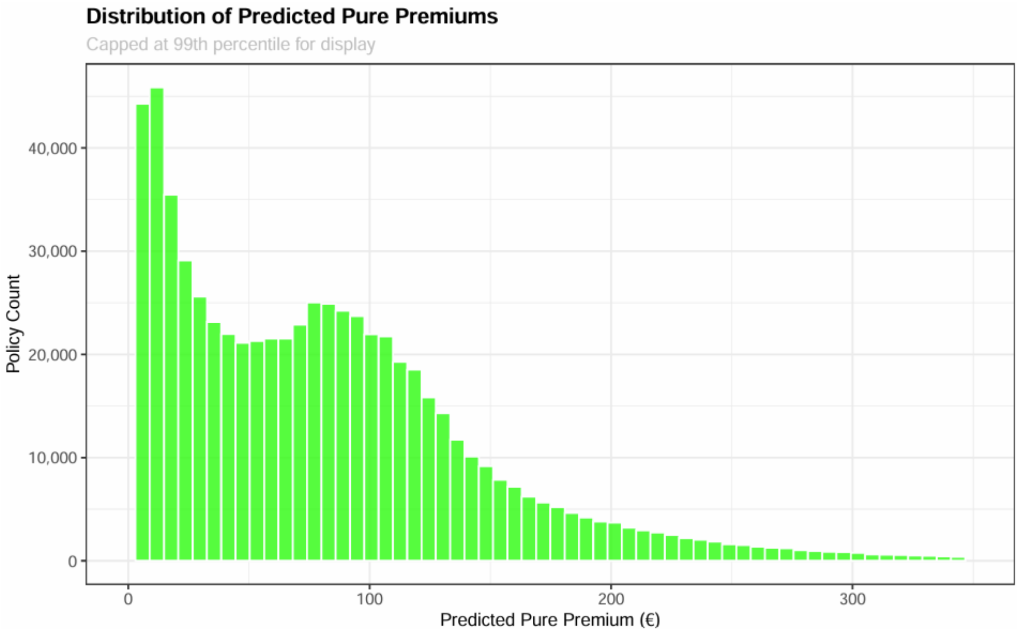


Figure 7

5.3.4. Conclusion

The pure premium model demonstrates good risk segmentation and excellent frequency calibration, with rating relativities that are consistent with expectations. However, the model materially overestimates aggregate loss costs, suggesting that the severity component requires further improvements. Despite this limitation, the resulting pure premium distribution appears realistic and provides meaningful differentiation between low- and high-risk policyholders, supporting its use as a pricing framework subject to further calibration of claim severity.

6. Executive Summary

A frequency-severity pricing framework was developed using Poisson and Gamma Generalised Linear Models on a Motor Third Party Liability portfolio. The final model identified Bonus-Malus score, driver age and vehicle age as the strongest predictors of expected claim costs. While the frequency model demonstrated good calibration and risk ranking ability, the severity model exhibited greater uncertainty and remains the primary area for future improvement. Overall, the resulting pure premium model produced sensible rating relativities and provides a useful demonstration of a standard actuarial pricing approach.

7. Potential Next Steps

- Include additional policy level information such as driver experience, annual mileage, vehicle usage information etc. to improve the predictive ability of the model.
- Investigate Negative Binomial distribution for Frequency GLM.
- Explore Tweedie model instead of separate frequency and severity models.
- Test combinations of variable and include the interaction terms that would improve model fit.
- Review and refine the current rating factor bandings to improve stability, credibility and interpretability
- Expand the dataset to include a larger number of policies, ideally with multiple underwriting years, to support more robust validation.